P 2.12 n-th stationary state ψ_n: $\hat{H}\psi_n = E_n \psi_n$, $E_n = \hbar\omega(n+\frac{1}{2})$

$$x = \sqrt{\frac{\hbar}{2m\omega}}\,(\hat{a}_+ + \hat{a}_-) \quad , \quad p = i\sqrt{\frac{\hbar m\omega}{2}}\,(\hat{a}_+ - \hat{a}_-)$$

$$\langle x\rangle = \sqrt{\frac{\hbar}{2m\omega}} \int dx\, \psi_n^* (\hat{a}_+ + \hat{a}_-)\psi_n = \sqrt{\frac{\hbar}{2m\omega}} \int dx\, \psi_n^* \left(\sqrt{n+1}\,\psi_{n+1} + \sqrt{n}\,\psi_{n-1}\right)$$

$= 0$ because $\int dx\, \psi_n^* \psi_m = 0$ for $n \neq m$.

$\langle p\rangle = 0$ for the same reason.

$$\langle x^2\rangle = \frac{\hbar}{2m\omega} \int dx\, \psi_n^* \left(\hat{a}_+\hat{a}_+ + \hat{a}_-\hat{a}_- + \hat{a}_+\hat{a}_- + \hat{a}_-\hat{a}_+\right)\psi_n$$

$$= \frac{\hbar}{2m\omega} \int dx\, \psi_n^* \Big(\sqrt{n+1}\sqrt{n+2}\,\psi_{n+2} + \sqrt{n}\sqrt{n-1}\,\psi_{n-2}$$

$$+ \sqrt{n}\,\hat{a}_+\psi_{n-1} + \sqrt{n+1}\,\hat{a}_-\psi_{n+1}\Big) =$$

$$= \frac{\hbar}{2m\omega} \int dx \left(\sqrt{n}\cdot\sqrt{n}\,\psi_n^*\psi_n + \sqrt{n+1}\sqrt{n+1}\,\psi_n^*\psi_n\right) = \frac{\hbar}{2m\omega}(n+n+1)$$

$$= \frac{\hbar}{2m\omega}(2n+1)$$

$$\langle p^2\rangle = -\frac{\hbar m\omega}{2} \int dx\, \psi_n^* \left(\hat{a}_+\hat{a}_+ + \hat{a}_-\hat{a}_- - \hat{a}_+\hat{a}_- - \hat{a}_-\hat{a}_+\right)\psi_n =$$

$$= -\frac{\hbar m\omega}{2} \int dx\, \psi_n^* \left(-n-(n+1)\right)\psi_n = \frac{\hbar m\omega}{2}(2n+1)$$

Then
$$\left.\begin{aligned} \sigma_x^2 &= \langle x^2\rangle - \langle x\rangle^2 = \frac{\hbar}{m\omega}\left(n+\frac{1}{2}\right) \\ \sigma_p^2 &= \langle p^2\rangle - \langle p\rangle^2 = \hbar m\omega\left(n+\frac{1}{2}\right) \end{aligned}\right\}$$

$\sigma_x \sigma_p = \hbar\left(n+\frac{1}{2}\right) \geq \frac{\hbar}{2}$ for all n, including the ground state: $n=0$.

P2.14 Initial state $\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}$: ground state of oscillator with frequency ω

Basis states: $\varphi_n(x)$: stationary states of oscillator with frequency $\omega' = 2\omega$.

We need to expand initial state ψ_0 in terms of φ_n:

$$\psi_0(x) = \sum_{n=0}^{\infty} c_n \varphi_n(x)$$

The lowest energy state is $\varphi_0(x)$: $E_0 = \hbar\omega' \cdot \frac{1}{2} = \hbar\omega$

The next is $\varphi_1(x)$: $E_1 = \hbar\omega'(1+\frac{1}{2}) = 3\hbar\omega$

$\Rightarrow$ probability to measure $\frac{1}{2}\hbar\omega$ is zero as there are no states with such energy.

Probability to measure $\hbar\omega$ is determined by c_0:

$P = |c_0|^2$.

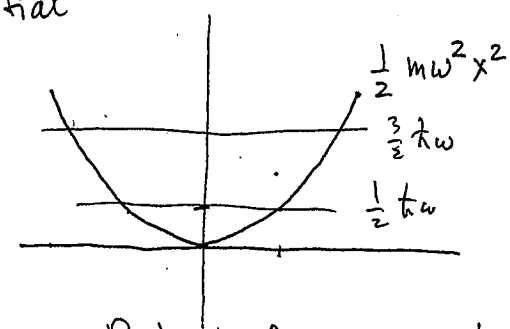
Find $c_0 = \int_{-\infty}^{\infty} dx\, \varphi_0^*(x)\, \psi_0(x) = \left(\frac{m\cdot 2\omega}{\pi\hbar}\right)^{1/4}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}$.

$$\cdot \int dx\, e^{-\frac{m2\omega}{2\hbar}x^2}\, e^{-\frac{m\omega}{2\hbar}x^2} = 2^{1/4}\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}.$$

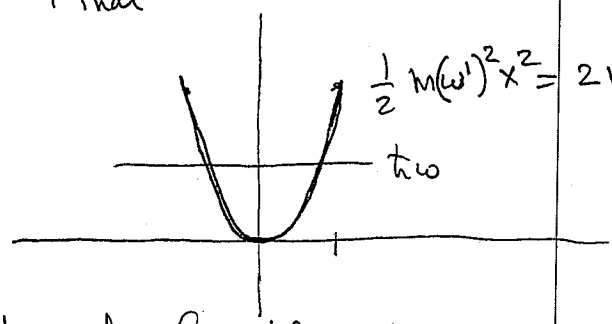
$$\cdot \int_{-\infty}^{\infty} dx\, e^{-\frac{3m\omega}{2\hbar}x^2} = 2^{1/4}\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\sqrt{\frac{\pi\cdot 2\hbar}{3m\omega}} = \sqrt{\frac{2}{3}}\cdot 2^{1/4}$$

$$\Rightarrow P = |c_0|^2 = \frac{2}{3}\cdot\sqrt{2} = 0.943$$

Initial



Final



<u>Note:</u> Potential energy has changed abruptly $\Rightarrow$ energy of the particle is not conserved!

P2.18

let $f(x) = A e^{ikx} + B e^{-ikx} = C \cos kx + D \sin kx$

Then $A(\cos kx + i \sin kx) + B(\cos kx - i \sin kx) =$

$$= (A+B)\cos kx + i(A-B)\sin kx = C\cos kx + D \sin kx$$

$$\Rightarrow \begin{cases} A+B = C \\ i(A-B) = D \end{cases} \Rightarrow A - B = -iD$$

$$\Rightarrow \begin{cases} A = \dfrac{C-iD}{2} \\ B = \dfrac{C+iD}{2} \end{cases}$$

P2.19

The probability current $J = \dfrac{i\hbar}{2m}\left(\psi \dfrac{\partial \psi^*}{\partial x} - \psi^* \dfrac{\partial \psi}{\partial x}\right)$

For $\psi_k(x,t) = A e^{i(kx - \frac{\hbar k^2}{2m}t)}$

$$\psi \frac{\partial}{\partial x}\psi^* = |A|^2 e^{ikx} \frac{\partial}{\partial x} e^{-ikx} = -ik|A|^2$$

$$\psi^* \frac{\partial}{\partial x}\psi = |A|^2 e^{-ikx} \frac{\partial}{\partial x} e^{ikx} = ik|A|^2$$

$$\Rightarrow J(x,t) = \frac{i\hbar}{2m}\left(-ik|A|^2 - ik|A|^2\right) =$$

$$= \frac{\hbar k}{m}|A|^2$$

Note that J is in the direction of momentum $\underline{p = \hbar k}$

Since classically $\frac{\hbar k}{m} = \frac{p}{m} = v$ (velocity),

$J(x,t)$ does describe the flow of probability.

P2.21 $\Psi(x,0) = A\, e^{-a|x|}$

(a) $$1 = A^2 \int_{-\infty}^{\infty} dx\, e^{-2a|x|} = 2A^2 \int_0^{\infty} dx\, e^{-2ax} =$$

$$= \frac{2A^2}{2a} \Rightarrow A = \sqrt{a}$$

(b) $$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx\, \Psi(x,0)\, e^{-ikx} = \sqrt{\frac{a}{2\pi}} \int_{-\infty}^{\infty} e^{-a|x| - ikx}\, dx$$

$$= \sqrt{\frac{a}{2\pi}} \left\{ \int_0^{\infty} e^{-ax-ikx}\, dx + \int_{-\infty}^{0} e^{+ax-ikx}\, dx \right\}$$

$$= \sqrt{\frac{a}{2\pi}} \int_0^{\infty} \left(e^{-ax-ikx} + e^{-ax+ikx} \right) dx \quad \xrightarrow{x' = -x} \quad \int_0^{\infty} e^{-ax'+ikx'}\, dx'$$

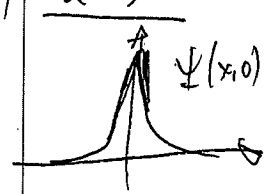
$$= \sqrt{\frac{a}{2\pi}} \left(\frac{1}{a+ik} + \frac{1}{a-ik} \right)$$

$$= \sqrt{\frac{a}{2\pi}}\, \frac{2a}{a^2+k^2}$$

(c) $$\Psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{\frac{a}{2\pi}}\, \frac{2a}{a^2+k^2}\, e^{i\left(kx - \frac{\hbar k^2}{2m} t\right)}\, dk$$

$$= \frac{a^{3/2}}{\pi} \int_{-\infty}^{\infty} \frac{e^{i\left(kx - \frac{\hbar k^2}{2m} t\right)}}{a^2+k^2}\, dk$$

(d) $\underline{a \to \infty}$ $\Rightarrow$ $\Psi(x,0) = \sqrt{a}\, e^{-a|x|}$ is very <u>sharply</u> peaked near $x=0$.



but $\phi(k) = \sqrt{\frac{2}{\pi}}\, \frac{a^{3/2}}{a^2+k^2} \to \sqrt{\frac{2}{\pi}}\, \frac{1}{\sqrt{a}}$ for $a \to \infty$ almost does not depend on $\underline{k}$ $\Rightarrow$ flat function!

$\underline{a \to 0}$ $\Rightarrow$ $\Psi(x,0) \cong \sqrt{a}$ (almost constant) but $\phi(k) \approx \sqrt{\frac{2}{\pi}}\, \frac{a^{3/2}}{k^2}$ is sharply peaked near $k=0$.

2.20

(a) $$f(x) = b_0 + \sum_{n=1}^{\infty} \frac{a_n}{2i}\left(e^{in\pi x/a} - e^{-in\pi x/a}\right) + \sum_{n=1}^{\infty} \frac{b_n}{2}\left(e^{\frac{in\pi x}{a}} + e^{-in\pi x/a}\right)$$

$$= b_0 + \sum_{n=1}^{\infty}\left(\frac{a_n}{2i} + \frac{b_n}{2}\right)e^{in\pi x/a} + \sum_{n=1}^{\infty}\left(-\frac{a_n}{2i} + \frac{b_n}{2}\right)e^{-in\pi x/a}$$

Choose

$$C_0 = b_0 \; ; \quad C_n = \frac{1}{2}(-ia_n + b_n), \text{ for } n = 1, 2, \ldots$$

$$C_n = \frac{1}{2}(-ia_{-n} + b_{-n}), \text{ for } n = -1, -2, -3, \ldots$$

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{-in\pi x/a}$$

(b) $$\int_{-a}^{a} f(x) e^{-im\pi x/a} dx = \sum_{n=-\infty}^{\infty} C_n \int_{-a}^{a} e^{i(n-m)\pi x/a} dx \; ; \text{ But for } n \neq m$$

$$\int_{-a}^{a} e^{i(n-m)\pi x/a} dx = \frac{e^{i(n-m)\pi x/a}}{i(n-m)\pi/a}\Bigg|_{-a}^{a} = \frac{e^{i(n-m)\pi} - e^{-i(n-m)\pi}}{i(n-m)\pi/a}$$

$$= \frac{(-1)^{n-m} - (-1)^{n-m}}{i(m-m)\pi/a} = 0$$

For $n = m \rightarrow \int_{-a}^{a} e^{i(n-m)\pi x/a} dx = \int_{-a}^{a} dx = 2a$

All terms except $n = m$ are zero;

and

$$\int_{-a}^{a} f(x) e^{-im\pi x/a} = 2a\, C_m \; ; \text{ hence } C_m = \frac{1}{2a}\int_{-a}^{a} f(x) e^{-\frac{im\pi x}{a}} dx$$

c

$$f(x) = \sum_{n=-\infty}^{\infty} \sqrt{\frac{\pi}{2}} \frac{1}{a} F(k) e^{ikx}$$

$$= \sum_{k=n} \frac{F(k) e^{ikx} \Delta k}{\sqrt{2\pi}}$$

where $\Delta k = \pi/a$ is the increment

in k from n to $(n+1)$

$$F(k) = \sqrt{\frac{2}{\pi}} \frac{1}{2a} \int_{-a}^{a} f(x) e^{-ikx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-a}^{a} f(x) e^{-ikx} dx$$

d As $a \to \infty$, k becomes a continuous variable,

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(k) e^{ikx} dk$$

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{ikx} dx.$$